

SAS Search - How to Use

This guide covers how to run the scripts, interpret the outputs, and use Helios Assist to extract the success definitions from the results.

All scripts are in: `sas_search/scripts/` All prompts are in: `sas_search/prompts/`

Getting Started - Setting Up VS Code

You need VS Code to open and run the scripts. If you already have it set up, skip to the next section.

Opening VS Code

1. Open VS Code from your Start menu or taskbar
2. Go to File > Open Folder and navigate to the Success Library folder on the network drive
3. On the left panel you'll see the file explorer - expand `sas_search/scripts/` to see all scripts

Opening the Terminal

You need a PowerShell terminal to run the scripts.

1. In VS Code, go to Terminal > New Terminal (or press `Ctrl+``)
2. A panel opens at the bottom of VS Code - this is your terminal
3. Make sure it says PowerShell in the top-right of the terminal panel. If it says something else (like "bash" or "cmd"), click the dropdown arrow next to the + button and select PowerShell
4. You're ready to run scripts

Running a Script

1. Open the script file in VS Code (click it in the left panel)
 2. Change the paths at the top (see each step below for details)
 3. Save the file (`Ctrl+S`)
 4. In the terminal, type the path to the script and press Enter:
 5. ```
 6. `.\sas_search\scripts\step1_scan.ps1`
 7. ```
 8. Or right-click the script file in the left panel and select Run in Terminal
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Quick Search - Single Mnemonic Lookup

If you need to look up one mnemonic (or a small handful), use `quick_search.ps1`. It does everything in one script - scan, tag, and extract - with no CSVs or pipeline steps.

Script: `sas_search/scripts/quick_search.ps1`

How to Use

1. Open the script in VS Code
2. Edit the three sections at the top:

```
# Your mnemonics (one or many)
$mnemonics = @("FTH", "MOM")

# Your keywords (searched IN ADDITION to "success" which is always included)
$keywords = @("mortgage", "open", "funded", "approved", "maturity", "retain")

# Paths
$searchPath = "\\maple.fg.rbc.com\data\Toronto\wrkgrp\..."
$outFile     = "\\maple.fg.rbc.com\data\...\quick_search_output.txt"
```

- 1. Save (Ctrl+S)
- 2. Run in terminal:
- 3. ``
- 4. .\sas_search\scripts\quick_search.ps1
- 5. ``

What It Does

Step	Action
1/3 Scan	Finds all .sas files on the network that contain y
2/3 Tag	For each file, records which of your mnemonics it
3/3 Extract	Pulls code blocks (20 lines above / 20 lines below

Output

A single text file with code blocks grouped by mnemonic. Each block is annotated with line markers:

Marker	Meaning
>>>	Line where "success" was found
>>>[keyword]	Line where one of your keywords was found (keyword
?	Line with conditional logic (if/where/case/when)
(blank)	Context line

Settings

These can be adjusted but usually don't need to be:

Setting	Default	What it controls
\$contextAbove	20	Lines to show above each keyword match
\$contextBelow	20	Lines to show below each keyword match
\$maxBlocks	10	Max code blocks per mnemonic-file pair

When to Use Quick Search vs the Pipeline

Scenario	Use
Looking up 1-3 mnemonics, want fast results	Quick Search
Running a batch of many mnemonics	Pipeline (step 1 - 4a)
Need to submit results to Helios Assist for AI ext	Pipeline (structured output with headers)
Need a summary dashboard of all mnemonics	Pipeline (step 5)

Quick Search is ideal for ad-hoc investigation - when you need to quickly see what SAS code exists for a specific campaign before deciding next steps.

The Pipeline

The default pipeline is:

Step 1 (scan) -> Step 2 (tag) -> Step 4a (extract) -> Split -> Helios Assist

Script	What it does	Input	Output
step1_scan.ps1	Finds all .sas files containing any mnemonic	Network folder + keyword_mapping.csv	CSV of matching files
step2_tag.ps1	Tags which mnemonics are in each file	Step 1 output + keyword_mapping.csv	CSV with mnemonic-file pairs
step4a_extract_success.ps1	Extracts code blocks around the word "suc	Step 2 output + keyword_mapping.csv	Text file with code extracts
split_extracts.ps1	Splits large text files for Helios	Any step 4 text output	Multiple text files (part1.txt, part2.txt, etc.)

If step 4a has gaps (mnemonics came back as NOISE or LABEL ONLY from Helios):

Script	What it does
step4b_extract_keywords.ps1	Fallback - extracts code blocks around mapped busi

Other tools:

Script	What it does
step5_summary.ps1	Dashboard CSV - status of every mnemonic
step3_dedup.ps1	Optional - narrows to one file per mnemonic (not u
step4_extract.ps1	Legacy - extracts around mnemonic reference (finds

Each script is standalone. You can run any one of them by itself as long as you have the input file it needs.

How to Edit Any Script

Every script has the same layout at the top - paths marked with === IN === and === OUT ===:

```
# === IN: ... ===
$inFile = "\\path\to\input.csv"

# === IN: mnemonic reference file ===
$mappingFile = "\\path\to\keyword_mapping.csv"

# === OUT: where results go ===
$outFile = "\\path\to\output.csv"
```

1. Open the script in VS Code
2. Change only the text between the quotes "..." for each path
3. Save (Ctrl+S)
4. Run the script

The output folder is created automatically if it doesn't exist.

Step 1 - Scan

Scans a network folder for .sas files that contain any mnemonic code. This is the slow step (network search).

Change these paths:

- \$searchPath - the root folder to search
- \$mappingFile - path to keyword_mapping.csv
- \$outFile - where to save the results

Output columns: FileName, FolderPath, FullPath

Step 2 - Tag

Opens each file from step 1 and records which specific mnemonics are inside it, when it was last modified, and whether it contains the

word "success". Also reports which mnemonics from keyword_mapping.csv were not found in any file.

Step 2 keeps all files per mnemonic - no deduplication. A mnemonic that appears in 5 different SAS files will have 5 rows.

Change these paths:

- \$inFile - the CSV from step 1 (or the v1 output: sas_mnemonic_files.csv)
- \$mappingFile - path to keyword_mapping.csv
- \$outFile - where to save the results

Output columns: Mnemonic, FileName, LastModified, FullPath, HasSuccess

Step 4a - Extract (default)

Searches for the word "success" in each SAS file. Pulls 20 lines above and 20 below each match. This targets the actual success measurement logic directly.

Since step 2 keeps all files per mnemonic, step 4a will extract from every file for each mnemonic. The same mnemonic may produce multiple sections in the output - one per file. The Helios prompts handle this by asking the AI to compare versions and pick the best one.

Change these paths:

- \$inFile - the CSV from step 2
- \$mappingFile - path to keyword_mapping.csv
- \$outFile - where to save the text file

Settings you can adjust:

- \$contextAbove - lines to show above each match (default: 20)
- \$contextBelow - lines to show below each match (default: 20)
- \$maxBlocks - max code blocks per mnemonic per file (default: 10)

Output markers:

Marker	Meaning
>>>	Line where "success" was found
?	Line with conditional logic (if/where/case/when)
(blank)	Context line

Step 4b - Extract (fallback)

Searches for mapped keywords specific to each mnemonic (from the Suggested_Keywords column in keyword_mapping.csv). For example, for FTH it searches for "mortgage, open, approved, funded, fhsa, homebuyer, close, booked".

Use this only for mnemonics where step 4a didn't find usable code - meaning the word "success" wasn't helpful but the actual logic might use business-specific terms instead.

Same paths and settings as step 4a.

Additional output marker:

Marker	Meaning
>>>[keyword]	Line where a mapped keyword was found (keyword nam

Split Script - split_extracts.ps1

Splits large step 4 output files into smaller parts that fit within Helios Assist's character limit. Splits on mnemonic section boundaries (never cuts a section in half).

Change these paths:

- \$inFile - the text file to split (any step 4a/4b output)
- \$outFolder - folder where parts go (created automatically)
- \$maxChars - max characters per part (default: 275,000)

Output: part1.txt, part2.txt, etc. in the output folder. If a single mnemonic section is larger than the limit, it gets saved as its own part with a warning.

Step 5 - Summary

Generates a dashboard CSV showing the status of every mnemonic: found or not, how many files, which have success logic.

Change these paths:

- \$inFile - the CSV from step 2
- \$mappingFile - path to keyword_mapping.csv
- \$outFile - where to save the summary

Output columns: Mnemonic, Description, Product, ExpectedSuccess, FileFound, FileCount, HasSuccessTerm, Files, LatestModified

Using Helios Assist to Extract Success Definitions

After running step 4a, you'll have a text file with SAS code extracts. The next step is to submit that file to Helios Assist to identify and extract the actual success code.

File Size Limit

Helios Assist has a limit of approximately 275,000 characters per submission. If your output file is larger than that, split it using split_extracts.ps1.

Submitting to Helios Assist

There are two phases. Phase 1 is a quick scan. Phase 2 is the detailed extraction.

The prompt files are in: sas_search/prompts/

Prompt file	Use with
prompt_broad_4a_success.md	Step 4a output (default - multiple files per mnemo
prompt_broad_4b_keywords.md	Step 4b output (fallback - multiple files per mnem
prompt_step4a_success.md	Step 4a output if you used optional step 3 dedup (
prompt_step4b_keywords.md	Step 4b output if you used optional step 3 dedup (

For the default pipeline (no step 3), use the broad prompts. These handle multiple files per mnemonic - they ask the AI to compare versions by file name and date, then pick the best one.

Recommended Workflow

1. Run the default pipeline: step 1 -> step 2 -> step 4a -> split if needed
2. Submit to Helios with prompt_broad_4a_success.md - run Phase 1
3. Review Phase 1 results. Note which mnemonics came back as LABEL ONLY, NOISE, or UNCLEAR

4. For those gaps: run step 4b for those mnemonics -> submit with prompt_broad_4b_keywords.md

Phase 1 - Scan and Categorize (do this first)

This is a quick pass to find out which mnemonics have usable code and which are noise.

1. Open the prompt file that matches your extraction type
2. Upload your output file (or paste the content) to Helios Assist
3. Copy the PHASE 1 prompt from the prompt file and paste it into Helios Assist
4. Submit

Helios will return a categorization table:

Mnemonic	File	Modified	Category	Note
FTH	prog_a.sas	2024-11-15	COMPLETE	Success defined as mortgage funded within
FTH	prog_b.sas	2023-06-20	NOISE	"mortgage" matched in a comment only
MOM	renew_v3.sas	2024-09-01	PARTIAL	Renewal flag visible but conditions cut off

Categories for step 4a output:

Category	Meaning	Action
COMPLETE	Full success logic is visible	Proceed to Phase 2
PARTIAL	Some logic visible but cut off	Proceed to Phase 2
LABEL ONLY	"success" is just a variable name, no logic	Try step 4b
UNCLEAR	Cannot determine	Try step 4b

Categories for step 4b output:

Category	Meaning	Action
USEFUL	Code clearly relates to success measurement	Proceed to Phase 2
PARTIAL	Some relevant logic but incomplete	Proceed to Phase 2
NOISE	Keywords matched but code is unrelated	Manual investigation needed
UNCLEAR	Cannot determine	Manual investigation needed

Phase 2 - Extract the Code (only for usable mnemonics)

This is where you get the actual SAS code - the asset we're building.

1. From the Phase 1 results, make a list of the mnemonics that came back as COMPLETE/USEFUL or PARTIAL
2. Copy the PHASE 2 prompt from the same prompt file
3. Replace [LIST THE COMPLETE AND PARTIAL ONES HERE] with your list of mnemonics
4. Submit to Helios Assist

Helios will return for each mnemonic:

- The expected success (from the header)
- The actual SAS code copied from the extract - this is the main deliverable
- A plain language summary of what the code does
- Whether the code matches the expected success
- Any gaps if the code is incomplete

Save the Phase 2 output. The SAS code it returns is the success definition asset for your library.

If the File is Too Large for One Submission

1. Split the file using split_extracts.ps1
2. Run Phase 1 on each part separately
3. Combine the Phase 1 tables from all parts
4. Run Phase 2 only for the usable mnemonics (you can combine them into one Phase 2 request if it fits)

Using Existing v1/v2 Files

These scripts are backwards compatible. If you already ran v1 or v2, you can skip steps:

You already have	Point it to	Then run from
sas_mnemonic_files.csv (v1 output)	Step 2 \$inFile	Step 2
sas_success_by_mnemonic.csv (v2 section 1)	Step 4a/4b \$inFile	Step 4a
sas_latest_per_mnemonic.csv (v2 section 2)	Step 4a/4b \$inFile	Step 4a

The HasSuccess column is optional - if your file doesn't have it, the script computes it automatically.

Adding New Campaigns

1. Open keyword_mapping.csv
2. Add a new row with the mnemonic, description, product, and expected success
3. Re-run from whatever step you need

The scripts read the mnemonic list from this file - no code changes needed.

Key columns in keyword_mapping.csv:

Column	What it controls
Mnemonic	The campaign code to search for in SAS files
Description	Shows in step 4 output header
Product	Shows in step 4 output header
Event_Type	Shows in step 4 output header
Event_Category	Shows in step 4 output header
Clean_Primary	Shows as EXPECTED SUCCESS in step 4 output
Primary_Subset	Shows as SUBSET/QUALIFIER in step 4 output
Suggested_Keywords	Used by step 4b to search for business terms in SA

All Files Reference

File	Location	What it is
keyword_mapping.csv	sas_search/	Master list of all campaigns - source of truth
keyword_mapping_final.csv	sas_search/	Campaigns grouped by product (reference only)
quick_search.ps1	sas_search/scripts/	All-in-one single mnemonic lookup (scan + tag + ex
step1_scan.ps1	sas_search/scripts/	Broad scan script
step2_tag.ps1	sas_search/scripts/	Tagging script (+ missing mnemonics report)
step4a_extract_success.ps1	sas_search/scripts/	Extract around "success" keyword (default)
step4b_extract_keywords.ps1	sas_search/scripts/	Extract around mapped keywords (fallback)
step5_summary.ps1	sas_search/scripts/	Summary dashboard
split_extracts.ps1	sas_search/scripts/	Split large text files for Helios Assist
step3_dedup.ps1	sas_search/scripts/	Optional - dedup to one file per mnemonic
step4_extract.ps1	sas_search/scripts/	Legacy - extract around mnemonic reference
prompt_broad_4a_success.md	sas_search/prompts/	Helios prompts for step 4a output (default)
prompt_broad_4b_keywords.md	sas_search/prompts/	Helios prompts for step 4b output (fallback)
prompt_step4a_success.md	sas_search/prompts/	Helios prompts for step 4a with optional dedup
prompt_step4b_keywords.md	sas_search/prompts/	Helios prompts for step 4b with optional dedup